DRG. NO. REF-001 · SHEET 1 / 4

REF/AR

SECTION A · THE HARDWARE

A domestic refrigerator, taken apart

An original 3D model of a household frost-free domestic refrigerator, exploded into its parts. Numbers match the legend.

Explore it in 3D
Scan for interactive web explorer & AR view.

DEPARTMENT OF ROBOTICS & AI, COEP TECH, PUNE
GUIDE: DR. SHUBHANGI DESHMUKH · B.TECH PROJECT

FIG. 1A · ASSEMBLED (OPEN DOORS)

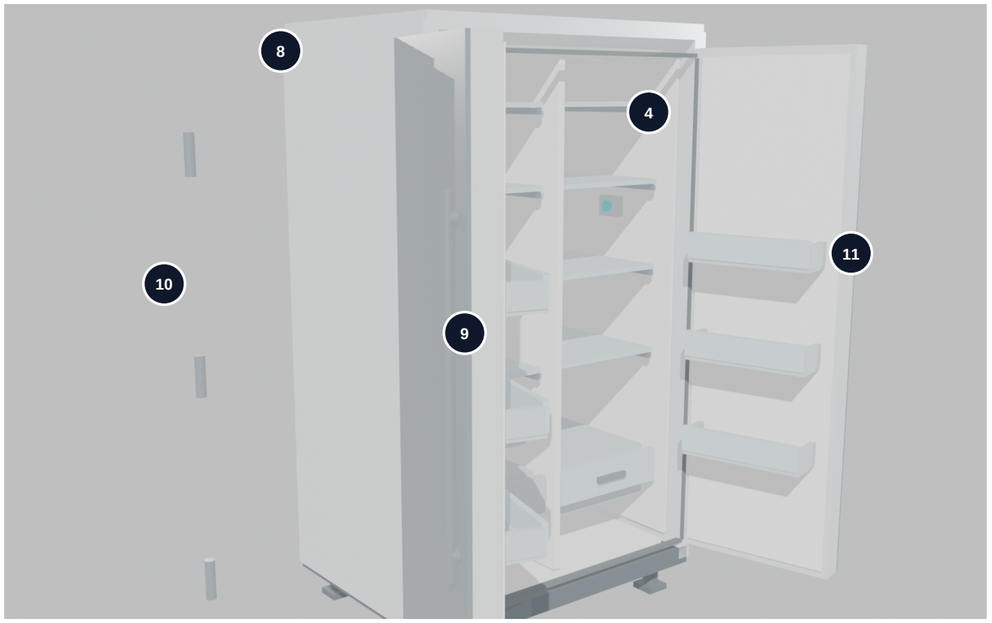
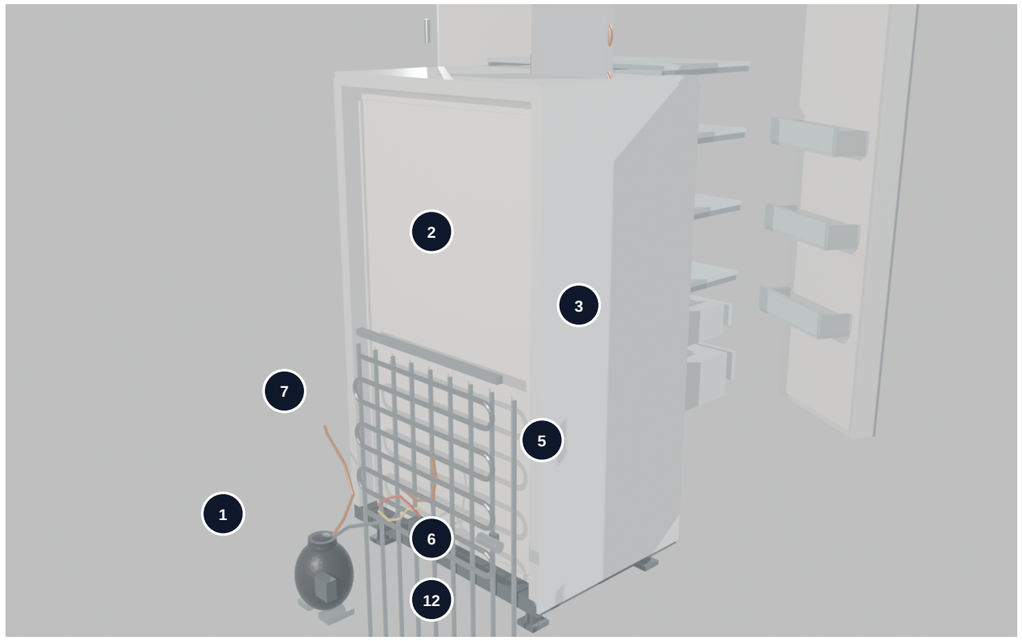


FIG. 1B · EXPLODED REAR VIEW



1. **Hermetic Compressor** — Draws in LP refrigerant vapour from evaporator, compresses it to HP gas.
2. **Condenser Coil** — Rejects heat (Q_out) to surroundings, condensing HP vapour to liquid.
3. **Capillary Tube** — Throttles high-pressure liquid, plunging pressure & saturation temperature.
4. **Evaporator Unit** — Absorbs heat (Q_in) from food space, vaporising sub-zero refrigerant.
5. **Filter-Drier** — Adsorbs trace moisture (zeolite beads) & filters debris before capillary.
6. **Condenser Fan** — Forces ambient room air across condenser coil and compressor casing.
7. **Refrigerant Circuit Lines** — Discharge, condenser return, and insulated suction lines.
8. **PUF Insulation & Cabinet** — Cyclopentane-blown foam (k ≈ 0.02 W/m·K) preventing heat leak.
9. **Shelves & Crisper Drawers** — Tempered glass shelves and humidity-regulated produce bins.
10. **Fresh Food Door Assembly** — Insulated left door with magnetic bellows gasket seal and bins.
11. **Freezer Door Assembly** — Sub-zero compartment right door with dual hinge pivot.
12. **Rear Ventilation Grille** — Chassis louvers channeling forced hot air exhaust from compressor.

Model data [VERIFIED GLTF 2.0]

MODEL:	Original procedural GLB, generated by generate_refrigerator_model.py
VALIDATION:	Khronos glTF-Validator: 0 errors, 0 warnings
HIERARCHY:	165 named nodes · 151 meshes · 15 PBR materials
DOOR HINGES:	Dual animated pivot axes (-0.485m & +0.485m hinge origins)
FOOTPRINT:	1.00 m wide × 0.72 m deep × 1.80 m tall (generic domestic scale)

Typical domestic nameplate specifications [REPRESENTATIVE REFERENCE]

REFRIGERANT:	R134a / R600a (Isobutane, GWP=3, ODP=0)
COMPRESSOR:	1/6 to 1/4 HP Hermetic Reciprocating (~120W to 180W)
CONDENSING P:	8.0 to 12.5 bar (~35°C to 45°C saturation temp)
EVAPORATING P:	1.1 to 1.4 bar (~ -20°C to -15°C saturation temp)
ESTIMATED COP:	1.8 to 2.4 (Typical seasonal household operating range)

Original procedural 3D geometry generated for engineering education. Verified glTF 2.0 asset. Renders from project Three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
THE HARDWARE: REFRIGERATOR, TAKEN APART	**Sahil Panchavishe 612302040** **Megh Shidam 612302058**	**Dr. Shubhangi Deshmukh** Dept of Robotics & AI · COEP Tech	**1 / 4** SCALE: NTS

REF/AR

DEPARTMENT OF ROBOTICS & AI, COEP TECH, PUNE
GUIDE: DR. SHUBHANGI DESHMUKH · B.TECH PROJECT

SECTION B · THE REFRIGERATION CYCLE

One loop, two pressures

How refrigerant carries heat out of a cold space: it boils at low pressure in the evaporator and condenses at high pressure in the condenser.

Explore it in 3D
Scan for interactive web explorer & AR view.

FIG. 2A · CYCLE SCHEMATIC

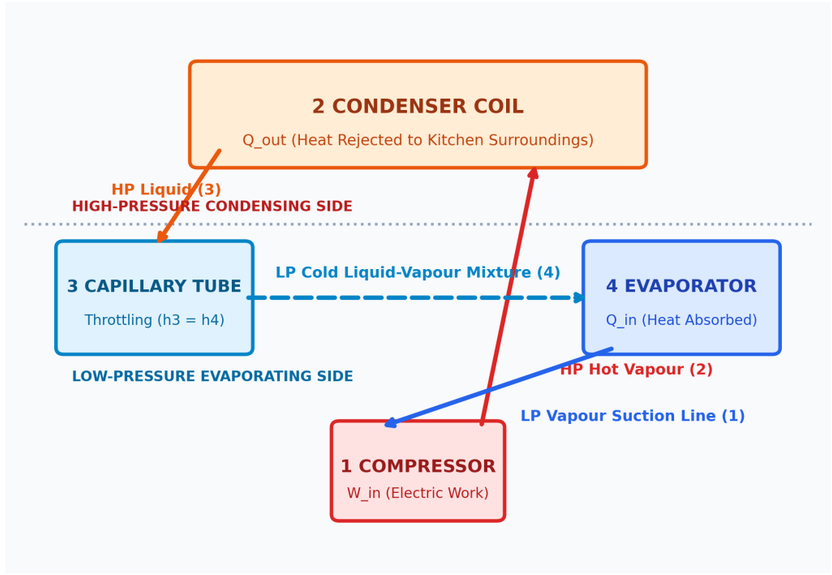
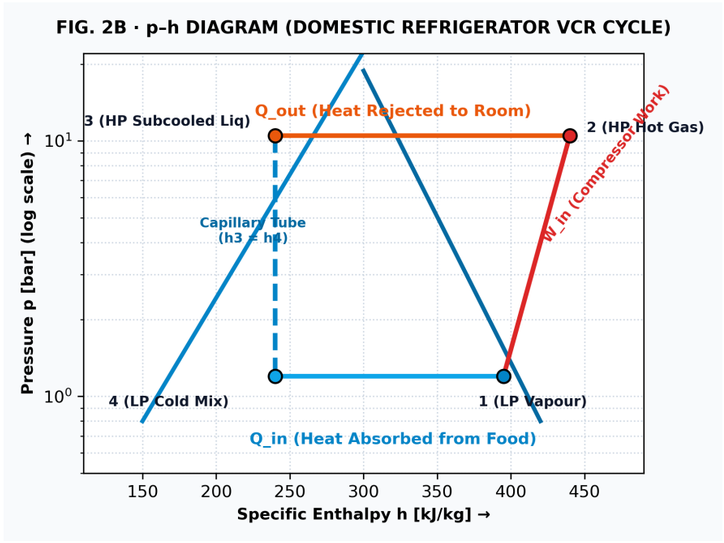


FIG. 2B · p–h DIAGRAM SKETCH



■ High-pressure side

Compressor outlet → condenser coil → filter-drier → capillary inlet. Refrigerant rejects heat (Q_out) here.

■ Low-pressure side

Capillary outlet → evaporator coil → suction line → compressor inlet. Refrigerant boils & absorbs heat (Q_in).

Energy balance (steady state):

$Q_{out} = Q_{in} + W_{in}$

$COP = Q_{in} / W_{in}$

FIG. 2C · FLOW TRACE IN 3D



1 Evaporation (heat absorbed)

Cold, low-pressure liquid-vapour mixture enters the evaporator coil.

It absorbs heat from food and air in the freezer and fresh food compartments, boiling into low-pressure vapour

Ideal cycle:

Ideal: Constant-pressure heat addition until refrigerant is saturated vapour.

Real system:

Real: Slight frictional pressure drop across coil; vapour is superheated 3-5 K at outlet to protect

2 Compression (work input)

Compressor draws in low-pressure vapour and squeezes it into hot gas.

Driven by electric motor work input. Pressure and saturation temperature rise above room ambient.

Ideal cycle:

Ideal: Reversible isentropic (adiabatic) compression (s1 = s2).

Real system:

Real: Valve losses, motor winding heat, and piston friction make compression non-isentropic (s2 > s1

3 Condensation (heat rejected)

Hot, high-pressure vapour discharges into the rear condenser coil.

Rejects heat (Q_out = Q_in + W_in) to room air. Refrigerant desuperheats, condenses, and subcools.

Ideal cycle:

Ideal: Constant-pressure heat rejection to saturated liquid state.

Real system:

Real: Slight subcooling (3-8 K) occurs at coil exit, ensuring 100% liquid enters capillary tube.

4 Expansion (throttling)

High-pressure liquid is forced through the narrow capillary tube.

High hydraulic resistance causes pressure to drop abruptly. A fraction flashes to vapour, chilling the stream.

Ideal cycle:

Ideal: Irreversible throttling at constant enthalpy (h3 = h4, zero work).

Real system:

Real: Adiabatic flow with critical choking; soldered to suction line to form internal heat exchanger

Original procedural 3D geometry generated for engineering education. Verified glTF 2.0 asset. Renders from project Three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
THE REFRIGERATION CYCLE: THERMODYNAMIC PRINCIPLE	Sahil Panchavishe 612302040 Megh Shidam 612302058	Dr. Shubhangi Deshmukh Dept of Robotics & AI · COEP Tech	2 / 4 SCALE: NTS

DRG. NO. REF-001 · SHEET 3 / 4

REF/AR

DEPARTMENT OF ROBOTICS & AI, COEP TECH, PUNE
GUIDE: DR. SHUBHANGI DESHMUKH · B.TECH PROJECT

SECTION C · THE PROJECT

One model, three ways in

Objective, architecture and implementation of the Domestic Refrigerator AR Explorer: one procedural model, three ways to learn from it.

Explore it in 3D
Scan for interactive
web explorer & AR view.

C1 Objective & motivation

Students learn the vapour compression cycle as a theoretical four-stage schematic, but a real domestic refrigerator is a complex machine with hidden coils, capillary lines, and foamed insulation.

This project bridges the gap: every component can be tapped to inspect its thermodynamic function, the door opens smoothly on dual hinge pivots, and the life-size refrigerator can be placed in AR on a real floor or table — with zero app installation.

- Identify the 4 core cycle components in 3D.
- Trace high-pressure and low-pressure sides.
- Walk around the virtual appliance in life-size AR.
- Relate physical layout to the p-h diagram.

C2 Implementation architecture

generate_refrigerator_model.py (Blender Python)

Domestic_Refrigerator.glb (glTF 2.0)

Web Explorer
React · Three.js

Unity 6 URP Player

A3 Poster PDF Generator

Single shared JSON dataset drives website, Unity app, and this poster for 100% agreement.

C3 How it works

3D MODEL
Procedural geometry (compressor shell, swept copper tubing, serpentine condenser, evaporator fins) with dual animated door hinges exported to glTF 2.0.

WEB EXPLORER
Interactive tap/click raycast picking, single-part highlight, ghosted isolate mode, explode slider, cycle flow pulse animation, and full component library modal.

AUGMENTED REALITY
Zero-install WebAR: WebXR & Scene Viewer on ARCore Android; AR Quick Look on iOS. Live camera overlay fallback on desktops & laptops.

UNITY 6 PROJECT
URP scenes: MainMenu, RefrigeratorExplorer, and AR Foundation scene for optional native builds.

C4 Applications & limitations

USE IT FOR
- Pre-lab briefing before thermal lab experiments
- Self-study, revision, and 3D exploration on phones
- Visualizing concealed internal cooling components
- Explaining refrigeration to non-specialists

KEEP IN MIND
- Educational visualization — not manufacturing CAD
- Illustrative generic domestic proportions
- Flow animation is conceptual, not CFD simulation
- No proprietary brand specifications invented
- AR tracking relies on device ARCore / ARKit

FIG. 3A · WEB EXPLORER (DESKTOP)

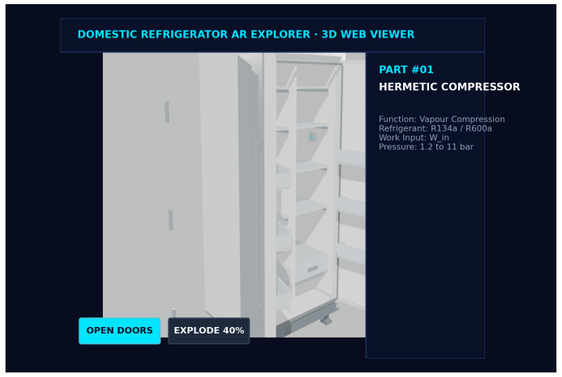


FIG. 3B · REAR MACHINERY ISOLATE



FIG. 3C · AR ROOM PLACEMENT

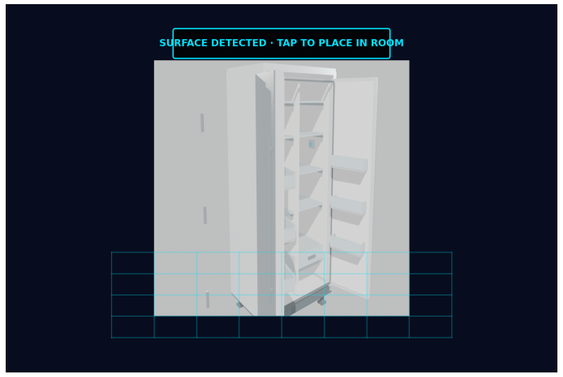


FIG. 3D · ASSEMBLED FRONT VIEW



Original procedural 3D geometry generated for engineering education. Verified glTF 2.0 asset. Renders from project Three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
THE PROJECT: HOW IT IS BUILT	**Sahil Panchavishe 612302040** **Megh Shidam 612302058**	**Dr. Shubhangi Deshmukh** Dept of Robotics & AI · COEP Tech	**3 / 4** SCALE: NTS

REF/AR

DEPARTMENT OF ROBOTICS & AI, COEP TECH, PUNE
GUIDE: DR. SHUBHANGI DESHMUKH · B.TECH PROJECT

SECTION D · ACCESS

Scan · Explore

Two codes, two ways in. Open your phone camera and point it at either code. Nothing to install.

Explore it in 3D
Scan for interactive
web explorer & AR view.

AUGMENTED REALITY

CODE 1 · AR

The fridge, in your room

Place the refrigerator on a real floor or table at life size, open the doors, and walk around it.

01 Scan, then tap View in your space.

02 Move the phone slowly over a floor or table.

03 Walk around it; tap Open Doors to see inside.

Android: Chrome with Google Play Services for AR (ARCore).
iPhone/iPad: Safari with AR Quick Look.
Camera permission asked only when AR starts.

WEB EXPLORER

CODE 2 · WEB

The whole appliance, interactive

Rotate the refrigerator in 3D, tap any of the 13 parts, open doors, and read the specifications.

01 Scan to open the 3D explorer in browser.

02 Tap compressor, condenser, valve or coils.

03 Toggle Open Doors or drag Explode slider.

Works in modern browsers on phones, tablets & laptops.
Nothing to install · Pure WebGL & WebXR.
Touch & mouse gestures for orbit, pan and zoom.

Project

TITLE	**Domestic Refrigerator AR Explorer — Interactive 3D & AR**
TEAM	**Sahil Panchavishe · 612302040** **Megh Shidam · 612302058**
BRANCH	**Robotics & AI (B.Tech)**
GUIDE	**Dr. Shubhangi Deshmukh**
INSTITUTION	**Department of Robotics & AI, COEP Technological University, Pune**

References & credits

- C. P. Arora, Refrigeration and Air Conditioning, 3rd ed., McGraw-Hill, 2008.
- W. F. Stoecker & J. W. Jones, Refrigeration and Air Conditioning, McGraw-Hill.
- Y. A. Çengel & M. A. Boles, Thermodynamics: An Engineering Approach, McGraw-Hill.
- ASHRAE Handbook — Refrigeration (Systems and Equipment), ASHRAE, 2022.
- Software: three.js, Google <model-viewer>, React 19, Vite 6, TypeScript, ReportLab.
- Unity: Unity 6 (URP, glTFast, AR Foundation, ARCore XR Plugin).
- 3D Model: Original procedural educational visualization created via Blender 5.2 Python API.
- Validation: Khronos Group glTF-Validator 2.0 (0 errors, 0 warnings).

Original procedural 3D geometry generated for engineering education. Verified glTF 2.0 asset. Renders from project Three.js and Blender engines.

TITLE	PREPARED BY · ROBOTICS & AI	GUIDED BY	SHEET
SCAN · EXPLORE: ACCESS & CREDITS	**Sahil Panchavishe 612302040** **Megh Shidam 612302058**	**Dr. Shubhangi Deshmukh** Dept of Robotics & AI · COEP Tech	**4 / 4** SCALE: NTS